JavaScript is disabled on your browser.

compact1, compact2, compact3

java.util

Class PropertyPermission

java.lang.Object
  java.security.Permission
    java.security.BasicPermission
      java.util.PropertyPermission

All Implemented Interfaces:

Serializable, Guard

```
public final class PropertyPermission
extends BasicPermission
```

This class is for property permissions.

The name is the name of the property ("java.home", "os.name", etc). The naming convention follows the hierarchical property naming convention. Also, an asterisk may appear at the end of the name, following a ".", or by itself, to signify a wildcard match. For example: "java.*" and "*" signify a wildcard match, while "*java" and "a*b" do not.

The actions to be granted are passed to the constructor in a string containing a list of one or more comma-separated keywords. The possible keywords are "read" and "write". Their meaning is defined as follows:

read

`read permission. Allows System.getProperty to be called.`

write

`write permission. Allows System.setProperty to be called.`

The actions string is converted to lowercase before processing.

Care should be taken before granting code permission to access certain system properties. For example, granting permission to access the "java.home" system property gives potentially malevolent code sensitive information about the system

environment (the Java installation directory). Also, granting permission to access the "user.name" and "user.home" system properties gives potentially malevolent code sensitive information about the user environment (the user's account name and home directory).

Since:

1.2

See Also:

BasicPermission, Permission, Permissions, PermissionCollection, SecurityManager

Constructor Summary

Constructors

Constructor	Description
PropertyPermission(String name, String actions)	Creates a new PropertyPermission object with the specified name.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
boolean	equals(Object obj)	Checks two PropertyPermission objects for equality.
String	getActions()	Returns the "canonical string representation" of the actions.
int	hashCode()	Returns the hash code value for this object.
boolean	implies(Permission p)	Checks if this PropertyPermission object "implies" the specified permission.

PermissionCollection	newPermissionCollection()	Returns a new PermissionCollection object for storing PropertyPermission objects.

Methods inherited from class java.security.Permission

checkGuard, getName, toString

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Constructor Detail

PropertyPermission

```
public PropertyPermission(String name,
                          String actions)
```

Creates a new PropertyPermission object with the specified name. The name is the name of the system property, and *actions* contains a comma-separated list of the desired actions granted on the property. Possible actions are "read" and "write".

Parameters:

`name - the name of the PropertyPermission.`

`actions - the actions string.`

Throws:

`NullPointerException - if name is null.`

`IllegalArgumentException - if name is empty or if actions is invalid.`

Method Detail

implies

```
public boolean implies(Permission p)
```

Checks if this PropertyPermission object "implies" the specified permission.

More specifically, this method returns true if:

- *p* is an instanceof PropertyPermission,
- *p*'s actions are a subset of this object's actions, and
- *p*'s name is implied by this object's name. For example, "java.*" implies "java.home".

Overrides:

`implies` in class `BasicPermission`

Parameters:

`p - the permission to check against.`

Returns:

`true if the specified permission is implied by this object, false if not.`

equals

`public boolean equals(Object obj)`

Checks two PropertyPermission objects for equality. Checks that *obj* is a PropertyPermission, and has the same name and actions as this object.

Overrides:

`equals` in class `BasicPermission`

Parameters:

`obj - the object we are testing for equality with this object.`

Returns:

`true if obj is a PropertyPermission, and has the same name and actions as this PropertyPermission object.`

See Also:

`Object.hashCode()`, `HashMap`

hashCode

`public int hashCode()`

Returns the hash code value for this object. The hash code used is the hash code of this permissions name, that is, `getName().hashCode()`, where `getName` is from the Permission superclass.

Overrides:

`hashCode` in class `BasicPermission`

Returns:

`a hash code value for this object.`

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

getActions

```
public String getActions()
```

Returns the "canonical string representation" of the actions. That is, this method always returns present actions in the following order: read, write. For example, if this PropertyPermission object allows both write and read actions, a call to `getActions` will return the string "read,write".

Overrides:

`getActions` in class `BasicPermission`

Returns:

the canonical string representation of the actions.

newPermissionCollection

```
public PermissionCollection newPermissionCollection()
```

Returns a new PermissionCollection object for storing PropertyPermission objects.

Overrides:

`newPermissionCollection` in class `BasicPermission`

Returns:

a new PermissionCollection object suitable for storing PropertyPermissions.

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms.
Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.